
Publication Decision from *Journal of High School Science*

"A Calibrated PCA–Machine Learning Pipeline for Biosignature Candidate Triage in Exoplanet Transmission Spectra"

Decision made on July 6, 2026

Editorial board's determination

Revise and resubmit

Comments from the editor

Dear author,

Please address all the comments of the reviewer and revise your manuscript accordingly.

When you resubmit, **please also submit separately a word file that lists verbatim each comment by the reviewer and then describe how and where in the revised manuscript you have addressed that particular comment. Do this for all comments in one separate file.**

Please note that if the the concerns/comments/questions of the reviewer are addressed inadequately, incompletely or insufficient address is paid to detail, your manuscript may be rejected. Please therefore thoroughly review your responses before resubmission.

We look forward to reviewing your revised manuscript.

Best,

Shireesh Apte

Reviewer 1

Rating scale questions

	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
The paper makes a significant contribution to scholarship.			✓		

	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
The literature review was thorough given the objectives and content of the article.			✓		

Open response questions

Comments to author

The manuscript presents a technically well-executed machine learning benchmark and the comparison among XGBoost, Random Forest, MLP, and CNN is carefully conducted. The statistical analyses are generally appropriate, the experimental pipeline is well documented, and the manuscript is clearly written. However, I believe the central scientific claims are presently insufficiently supported. The primary issue is not the implementation of the machine learning algorithms themselves but rather whether the experiments actually demonstrate what the manuscript claims. In its current form, the work demonstrates that an XGBoost classifier can successfully recover predefined biosignature labels from spectra generated within a single synthetic simulation framework. It does not yet demonstrate that the proposed methodology identifies biosignatures in actual exoplanet transmission spectra or that the mechanistic interpretation of the PCA representation is correct. These issues require additional experimental validation before the manuscript is suitable for publication.

The most significant concern is that the entire study is developed, trained, validated, and tested exclusively within a single synthetic ecosystem. Every transmission spectrum is generated using the TauREx/MultiREx forward model, every class label is derived directly from the same simulated atmospheric parameters, and every evaluation is performed on additional spectra generated under essentially identical modeling assumptions. Consequently, the classifier is never evaluated outside the statistical distribution of the simulator that created both the inputs and the labels. High predictive accuracy under these conditions demonstrates internal

consistency within the simulation environment but does not establish that the learned decision boundary represents genuine atmospheric physics rather than simulator-specific statistical regularities. The central question for a biosignature classifier is not whether it generalizes to additional TauREx simulations, but whether it generalizes to real exoplanet observations. The manuscript does not presently address this question.

This distinction is especially important because real Ariel observations will differ substantially from the synthetic spectra used in this work. Ariel will observe heterogeneous transmission spectra containing instrumental systematics, wavelength-dependent detector effects, correlated noise, stellar contamination, imperfect calibration, incomplete molecular opacity databases, cloud and haze variability, photochemical complexity, and three-dimensional atmospheric structure. In contrast, the present study assumes one-dimensional atmospheres, Gaussian noise with a fixed signal-to-noise ratio, predefined atmospheric compositions, and idealized observing conditions. These assumptions substantially reduce the complexity of the classification problem. Before this work can be considered applicable to Ariel, the authors should demonstrate that the classifier remains robust under realistic domain shift. At minimum, the manuscript should include experiments using spectra generated by independent radiative transfer codes, alternative molecular opacity databases, different cloud and haze prescriptions, different stellar contamination models, varying spectral resolutions and signal-to-noise ratios, more realistic instrumental systematics, and domain-shift or transfer-learning experiments. Without such validation, the manuscript demonstrates performance only within a single synthetic environment rather than on the class of observations for which the framework is ultimately intended.

A related concern is the definition of the positive class. The manuscript defines biosignatures using fixed abundance thresholds ($\text{CH}_4 > 10^{-6}$ and $\text{O}_3 > 10^{-7}$), and every atmosphere satisfying these thresholds is labeled as positive. Consequently, the classifier is not discovering biosignatures but rather learning a predefined labeling rule imposed during simulation. This distinction is important because Ariel itself does not operate by measuring atmospheric abundances and applying deterministic thresholds. Ariel measures transmission spectra, after which atmospheric abundances must be inferred through retrieval algorithms that incorporate observational uncertainties, parameter degeneracies, stellar context, and competing abiotic explanations. The proposed machine learning framework therefore

learns to reproduce a synthetic labeling convention rather than the operational biosignature identification process that Ariel will ultimately employ. Additional experiments using retrieval-derived abundances with realistic uncertainties or probabilistic labels would substantially strengthen the relevance of the proposed framework.

I also have substantial concerns regarding the interpretation of the PCA representation. The manuscript repeatedly argues that the first principal components primarily encode broad physical structure while higher-order components encode chemically informative absorption features. However, this interpretation is not supported by the mathematics of principal component analysis. Principal components are orthogonal linear combinations of all original wavelength variables. They are not independent physical variables, nor do they uniquely correspond to individual atmospheric processes. Every principal component generally contains different weighted mixtures of continuum structure, planetary radius, atmospheric temperature, molecular absorption, scattering, clouds, and measurement noise. Consequently, methane and ozone information is expected to be distributed across multiple principal components rather than residing exclusively within higher-order components. Demonstrating that PCo correlates strongly with mean transit depth does not imply that chemically relevant information is absent from that component, nor does it establish that chemistry has been isolated into PCs 2–101. These interpretations require considerably stronger evidence than presently provided. The manuscript should include loading-vector analyses, variance decomposition by physical parameter, reconstruction studies after selectively removing principal components, and explicit quantification of how methane and ozone absorption features project onto individual principal components before drawing these mechanistic conclusions.

This concern becomes even more significant following the whitening procedure. After PCA, every retained principal component is standardized to unit variance before training the neural network models. While this may improve optimization for gradient-based learning, it fundamentally changes the geometry of the feature space. Whitening does not selectively amplify chemically informative features. Instead, it amplifies every low-variance principal component equally, regardless of whether that component contains molecular absorption features, continuum information, numerical artifacts, simulator-specific structure, or measurement noise. Since each principal component is itself a linear combination of all original

wavelengths, whitening rescales mixtures of physical and chemical information rather than independently equalizing “physical” and “chemical” features. Consequently, the mechanistic explanation presented in the manuscript—that whitening specifically increases the influence of chemically informative components—is not mathematically demonstrated.

More fundamentally, the manuscript appears to contain a conceptual inconsistency. It argues that PCA naturally orders the data such that dominant physical information resides in the first principal components while chemically relevant information occupies lower-variance components. However, the subsequent whitening operation deliberately removes precisely this variance hierarchy by forcing every retained component to have identical variance. If the PCA variance ordering reflects meaningful atmospheric physics, it is unclear why the subsequent learning algorithm should deliberately eliminate that ordering. Conversely, if equalizing all component variances improves classification, this suggests that explained variance itself may not correspond to discriminative importance for biosignature detection. These two interpretations cannot simultaneously serve as the mechanistic explanation for the reported performance. The manuscript should explicitly resolve this conceptual inconsistency through additional ablation studies comparing whitening, removal of leading principal components, supervised dimensionality reduction techniques such as Linear Discriminant Analysis or Partial Least Squares, and alternative feature weighting strategies. These experiments are necessary to determine whether whitening is compensating for limitations of PCA or whether PCA is genuinely isolating chemically meaningful structure.

An equally important concern is that the whitening procedure may substantially reduce robustness to real observational data even if it improves performance within the synthetic simulation environment. Within TauREx, whitening appears to enhance classification by increasing the influence of low-variance components that contain discriminative methane and ozone information. However, because the entire pipeline is trained and evaluated on spectra generated by the same simulator, there is no evidence that these low-variance components represent exclusively chemically informative features. In real telescope observations, low-variance principal components are also expected to contain detector artifacts, calibration residuals, correlated instrumental noise, stellar contamination, imperfect molecular opacity models, and other observational effects that are absent or greatly simplified in the simulations. Whitening cannot distinguish

chemically meaningful variance from these alternative sources of variability; it amplifies all low-variance components equally. Consequently, a preprocessing strategy that improves classification by amplifying subtle methane and ozone signatures in synthetic data may equally amplify observational noise and instrumental artifacts when deployed on real Ariel spectra. The present manuscript provides no evidence that whitening enhances physically meaningful biosignature information rather than simulator-specific variance. Addressing this issue requires experiments on independent simulation environments and, where possible, observational spectra to determine whether whitening genuinely improves out-of-distribution generalization rather than simply increasing performance within the synthetic training distribution.

Overall, I believe the manuscript contains a promising machine learning framework and an extensive computational evaluation. However, the central scientific questions remain unresolved. Specifically, the manuscript has not yet demonstrated that the classifier generalizes beyond the synthetic TauREx environment, that the PCA interpretation accurately reflects the underlying atmospheric physics, or that the whitening procedure preferentially enhances chemically meaningful information rather than simulator-specific low-variance structure. These are not matters of presentation but of experimental validation. Accordingly, I recommend major revision. Before the manuscript can be considered for publication, the authors should perform the additional validation experiments outlined above, including independent simulation studies, realistic domain-shift analyses, robustness testing under observational systematics, mechanistic analyses of the PCA representation, and targeted ablation studies of the whitening procedure. These experiments are necessary to establish that the reported performance reflects robust biosignature detection rather than successful classification within a single internally consistent synthetic simulation framework.

Rating scale questions

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The paper makes a significant contribution to scholarship.			✓		
The literature review was thorough given the objectives and content of the article.			✓		

Open response questions

Comments to author

Please verify that all links to the references point to the correct source.

The authors must disclose and acknowledge any assistance received in the preparation of this manuscript, including but not limited to editorial, technical, analytical, or writing support. All such contributions must be clearly stated in the Acknowledgments section.

Include enough recent references along with foundational ones. Ensure references directly support your claims. Avoid “padding.” Use credible sources (peer-reviewed journals, reputable books, official reports).

All assumptions (including implicit assumptions) must be explicitly stated and clearly justified. The authors should explain why each assumption is reasonable and discuss its impact on the results and conclusions.

Avoid overreaching conclusions that extend beyond what is supported by the data and analysis.

Verify that you have used past perfect tense and third person throughout the manuscript wherever applicable.
